Queen Mary
University of London

EBU5376
Digital Signal Processing

Discrete Fourier Transform (DFT)

Module Organizer: Dr Yixuan Zou yixuan.zou@qmul.ac.uk

Outline

- Discrete Fourier Transform (DFT)
  - Duality in Fourier Transform
  - Twiddle Factors
  - Zero Padding
  - Properties of DFT
- Circular Convolution using DFT

Outline

- Discrete Fourier Transform (DFT)
  - Duality in Fourier Transform
  - Twiddle Factors
  - Zero Padding
  - Properties of DFT
- Circular Convolution using DFT

Problem with CTFT and DTFT

Discrete time Fourier transform (DTFT): $X(\omega) = \sum_{-\infty}^{\infty} x[t]e^{-j\omega t}$

Continuous time Fourier transform (CTFT): $X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$

- Both CTFT and DTFT require sampling the signal for an infinite number of times and produce a continuous spectrum, which requires evaluating infinitely many frequency points. **This is not feasible in practice**.
- Hence, in practical implementations, the Fourier transform is approximated by **finite samples in both the time and frequency domain**

Discrete Fourier Transform (DFT)

- **Duality in Fourier Transform**: Previously, we established that sampling in one domain results in periodicity in the other domain.
- **Time-Domain Sampling → Periodic Frequency Representation**:
  - In DFT, we take a **finite** number of time-domain samples.
  - This results in a **periodic spectrum** in the frequency domain.
- **Frequency-Domain Sampling → Periodic Time Representation**:
  - Since DFT considers only **a finite number of frequency samples** (also called **frequency bins**), it implicitly assumes the signal is **periodic in time**.
  - This periodic extension means the finite-time sequence repeats indefinitely.

Discrete Fourier Transform

DTFT: $$X(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}, \qquad \omega \in [-\pi, \pi]$$

- In DFT, the frequency variable is discretized to K equally spaced points:

$$\omega_k = \frac{2\pi k}{K}, \qquad k = 0, \dots, K-1$$

- Let N denotes the number of time-domain samples.
- The Discrete Fourier transform (DFT) can be formulated as

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-\frac{j2\pi kn}{N}}, \qquad k = 0, \dots, K-1$$

Magnitude of the kth frequency

Value of the nth sample

Discrete Fourier Transform

- N: Number of time-domain samples
  - Defines the length of the discrete signal $x[n]$
  - The signal is implicitly assumed to be periodic with period N
- K: Number of frequency bins (for a K-point DFT)
  - Defines the number of discrete frequency components we compute in the DFT
  - Each $X[k]$ represents the Fourier coefficient at frequency $\omega_k = \frac{2\pi k}{K}$
  - K controls frequency resolution: larger K provides finer frequency spacing $\Delta\omega = \frac{2\pi}{K}$.
  - The DFT spectrum is periodic, repeating every K in the frequency index and every 2π in normalized frequency.
  - To fully describe an N-sample discrete-time signal, the minimum number of frequency bins needed is $K = N$

Discrete Fourier Transform

DTFT: $X(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$

- The Discrete Fourier transform (DFT) is formulated as:

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-\frac{j2\pi kn}{N}}, \qquad k = 0, \dots, N-1$$

Magnitude of the kth frequency

Value of the nth sample

- The inverse DFT (IDFT) is formulated as

$$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]e^{\frac{j2\pi kn}{N}}, \qquad n = 0, \dots, N-1$$

- By default, the number of frequency bins is equal to the number of time-domain samples, i.e., $K = N$

Discrete Fourier Transform

- Simplified DFT formulation based on twiddle factor $W_N^{kn} = e^{-\frac{j2\pi kn}{N}}$

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-\frac{j2\pi kn}{N}} \quad \overset{DFT}{\longleftrightarrow} \quad x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k] e^{\frac{j2\pi kn}{N}}$$

$$W_N^{kn} = e^{-\frac{j2\pi kn}{N}}$$

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn} \quad \overset{DFT}{\longleftrightarrow} \quad x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k] W_N^{-kn}$$

- The twiddle factor W_N^{kn} has three parameters: 1) number of samples N; 2) Index of time-domain sample n; 3) index of frequency component k

Discrete Fourier Transform – Periodicity Assumption

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn} \quad \overset{DFT}{\longleftrightarrow} \quad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn}$$

- Some notes on DFT
  - We denote the DFT pair of $x[n]$ as $x[n] \overset{DFT}{\longleftrightarrow} X[k]$
  - Recall: The frequency spectrum of DTFT needs to be interpret in terms of the sampling frequency
  - Same applies to DFT.
  - The kth DFT coefficient $X[k]$ indicates the amplitude of the kth frequency
  - The kth frequency in a N-point DFT is $\frac{k}{N} f_s$ Hz ($= \frac{k}{N} 2\pi f_s$ radians), where f_s is the sampling frequency used to obtain the N time-domain samples.

Discrete Fourier Transform

- Example:

1. List all the distinct twiddle factors W_N^{kn} involved in calculating a 4-point DFT.
2. Then, evaluate each twiddle factor using the formula $W_N^{kn} = e^{-\frac{j2\pi kn}{N}}$.

Discrete Fourier Transform

- Example: 4-point DFT of a 4-point rectangular sequence

$$R_4[n] = \begin{cases} 0, & n < 0, n > 4 \\ 1, & 0 \leq n \leq 3 \end{cases}$$

Discrete Fourier Transform

- We can improve the frequency resolution of DFT via two schemes:
  1. **Increase N:** Collect more time-domain samples.
  2. **Zero-padding**: Extend the signal length by appending zeros before computing the DFT.

Discrete Fourier Transform

- **Zero-padding**
  - Append four number of zeros on the tail of $R_4[n]$ and we can get an 8-point sequence $[1,1,1,1,0,0,0,0]$
  - Then, the 8-point DFT can be obtained as

$$X_8[k] = \sum_{n=0}^{7} x[n] W_8^{kn}, \qquad k = 0, \dots, 7$$

  - The 8-point DFT results in frequency components at:

$$\left\{0, \frac{f_s}{8}, \frac{2f_s}{8}, \frac{3f_s}{8}, \frac{4f_s}{8}, \frac{5f_s}{8}, \frac{6f_s}{8}, \frac{7f_s}{8}\right\}$$

  - By computing the 8-point DFT for the rectangular sequence $R_4[n]$ we get

$$|X_8[k]| = [4, 2.61, 0, 1.08, 0, 1.08, 0, 2.61]$$

  - The 8-point DFT samples the DTFT at twice as many points, revealing values that were always present in the DTFT but unobserved in the 4-point DFT.

Summary on Zero-Padding

- When the standard N-point DFT provides an insufficient number of frequency bins to analyze the spectrum in detail, zero-padding increases the number of computed bins, making spectral peaks easier to identify
- If the input sequence has N samples and we compute a K-point DFT where $K > N$, appending $K - N$ zeros to the sequence is the standard way to perform the computation.
- When zero-padding is applied, the additional frequency bins correspond to more closely spaced samples of the DTFT.
- Although zero-padding provides a denser frequency representation, actual resolution improvement for distinguishing closely spaced frequency components requires a longer time-domain signal

Properties of DFT

- Linearity

$$ax_1[n] + bx_2[n] \overset{DFT}{\longleftrightarrow} aX_1[k] + bX_2[k]$$

- Time-shift

$$x[n-m] \overset{DFT}{\longleftrightarrow} W_N^{km} X[k]$$

- Frequency-shift

$$W_N^{-mn} x[n] \overset{DFT}{\longleftrightarrow} X[k-m]$$

Derive the proofs as an exercise

- DFT inherits many of the properties of DTFT.
- Listed above are the three basic and important properties associated with the Fourier transform methods, including DFT, DTFT, CTFT.

Properties of DFT

- **Periodicity** of N-point DFT

$$X[k] = X[k + mN]$$

- DFT inherits the periodicity property of DTFT
- It implies that the frequency spectrum of DFT repeats itself with a period of 2π
- In terms of the twiddle factors, the periodicity of DFT is expressed by

$$W_N^{(k+mN)n} = W_N^{kn}, \qquad \forall m \in \mathbb{Z}$$

$$W_N^{kn} = W_N^{(kn-N)}, \qquad if\ kn \geq N$$

- This is an important property that allows for the derivation of Fast Fourier transform (FFT)

Properties of DFT

➢ Convolution property of DFT

The DFTs are multiplied element-wise

$$x_1[n] * x_2[n] \overset{DFT}{\longleftrightarrow} X_1[k]X_2[k]$$

- ❖ The convolution property is shared by all Fourier transform methods, including DFT, DTFT, and CTFT
- ❖ For DFT, the input sequences are assumed to be periodic
- ❖ In terms of convolution, when the two input sequences are periodic with the same period, their linear convolution is equivalent to their circular convolution
- ❖ Hence, the convolution property of DFT can also be expressed as

$$x_1[n] \circledast x_2[n] \overset{DFT}{\longleftrightarrow} X_1[k]X_2[k]$$

Outline

- Discrete Fourier Transform (DFT)
  - Duality in Fourier Transform
  - Twiddle Factors
  - Zero Padding
  - Properties of DFT
- Circular Convolution using DFT

Circular Convolution using DFT

- Why the convolution property of DFT is important? One reason is efficiency.
- If we are dealing with long signals, then convolution is more computational intensive than multiplication.
- For circular convolution

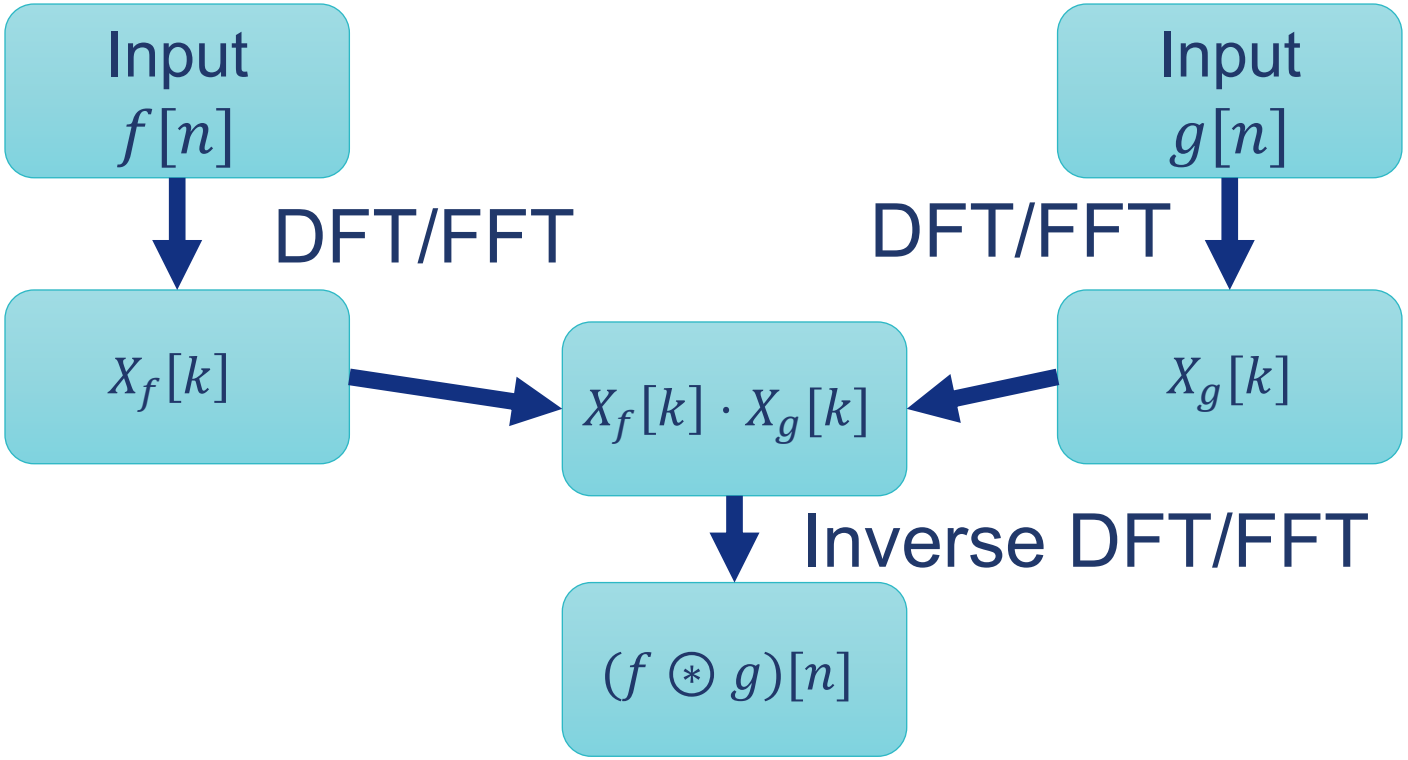
$$(f \circledast g)[n] = \sum_{m=0}^{N} f[\langle n-m \rangle_N] g[m]$$

- But for multiplication this is much simpler

$$(f \circledast g)[n] = \mathcal{F}^{-1}\left(X_f[k] \cdot X_g[k]\right)$$

- Especially for the circular convolution, we have Fast Fourier Transform to take advantage of

Circular Convolution using DFT



Circular Convolution using DFT

- Two sequences: $x[n] = [2, 2, 2]$ and $h[n] = [1, 1, 1, 1]$
- Compute the circular convolution $x[n] \circledast h[n]$ using DFT.

Summary – Fourier Family

- The family of Fourier transform:

1. Fourier series (FS) $$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos \omega n t + \sum_{n=1}^{\infty} b_n \sin \omega n t$$
2. Continuous time Fourier transform (CTFT) $$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$
3. Discrete time Fourier transform (DTFT) $$X(\omega) = \sum_{n=-\infty}^{n=\infty} x[n] e^{-j\omega n}$$
4. Discrete Fourier transform (DFT) $$X[k] = \sum_{n=0}^{N-1} x[n] e^{-\frac{j2\pi kn}{N}}$$

Summary – Fourier Family

- The family of Fourier transforms investigates how a signal can be decomposed into sine waves and cosine waves of different frequencies
- Both *FS* and *DFT* assume the signal to be periodic, whereas *CTFT* and *DTFT* can also be applied to aperiodic signals
- *FS, CTFT* and *DTFT* deal with infinite signals, *DFT* deals with finite signals
- *DFT* is the only one that can be implemented by computers

Summary – Fourier Family

	Time domain	Frequency domain
Fourier Series	Continuous, periodic, Infinite	Discrete, aperiodic, Infinite
Fourier Transform	Continuous, aperiodic, Infinite	Continuous, aperiodic, Infinite
Discrete Time Fourier Transform	Discrete, aperiodic, Infinite	Continuous, periodic, Infinite
Discrete Fourier Transform	Discrete, periodic, Finite	Discrete, periodic, Finite

Table: Comparing the time/frequency domain of the Fourier family

Queen Mary
University of London